

Layout Pages Practice

1. Kreiranje projekta

`npm create vite@latest layout-pages-practice -- --template react`

Uđemo u projekt:

`cd layout-pages-practice`

Instaliramo pakete:

`npm install`

Instaliramo React Router:

`npm install react-router-dom`

Pokrenemo aplikaciju:

`npm run dev`

2. Struktura projekta

Nakon čišćenja početnog Vite projekta, struktura treba izgledati ovako:

`src/`

`App.jsx`

`App.css`

`main.jsx`

`layout/`

`RootLayout.jsx`

`pages/`

`Home.jsx`

`Lessons.jsx`

`Resources.jsx`

`About.jsx`

`NotFound.jsx`

3. src/main.jsx

U većini Vite projekata ova datoteka je već spremna. Treba izgledati ovako:

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import App from "./App.jsx";

createRoot(document.getElementById("root")).render(
  <StrictMode>
    <App />
  </StrictMode>
);
```

Ovdje ne dodajemo router. Router ćemo staviti u `App.jsx`, zato što je to za sada jednostavnije i jasnije za učenje.

4. src/App.jsx

Ovdje definiramo sve rute aplikacije.

```
import { BrowserRouter, Route, Routes } from "react-router-dom";
import RootLayout from "../layout/RootLayout";
import Home from "../pages/Home";
import Lessons from "../pages/Lessons";
import Resources from "../pages/Resources";
import About from "../pages/About";
import NotFound from "../pages/NotFound";
import "../App.css";
```

```
function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<RootLayout />} />
        <Route index element={<Home />} />
        <Route path="lessons" element={<Lessons />} />
        <Route path="resources" element={<Resources />} />
        <Route path="about" element={<About />} />
        <Route path="*" element={<NotFound />} />
      </Routes>
    </BrowserRouter>
  );
}
```

```
);  
}
```

export default App;

Objašnjenje

Ovo je glavna ruta:

```
<Route path="/" element={<RootLayout />} />
```

Ona kaže React Routeru:

“Za sve ove child stranice koristi **RootLayout** kao zajednički okvir.”

Unutar nje imamo child rute:

```
<Route index element={<Home />} />  
<Route path="lessons" element={<Lessons />} />  
<Route path="resources" element={<Resources />} />  
<Route path="about" element={<About />} />  
<Route path="*" element={<NotFound />} />
```

index ruta znači:

Ako je korisnik točno na /, prikaži Home.

Ruta:

```
<Route path="lessons" element={<Lessons />} />
```

znači:

Ako je korisnik na /lessons, prikaži Lessons.

A fallback ruta:

```
<Route path="*" element={<NotFound />} />
```

znači:

Ako nijedna ruta ne odgovara, prikaži NotFound.

5. **src/layout/RootLayout.jsx**

Sada radimo najvažniju datoteku ove vježbe.

```

import { NavLink, Outlet } from "react-router-dom";

function RootLayout() {
  return (
    <div className="site-shell">
      <header className="site-header">
        <div className="brand">React Learning Studio</div>

        <nav className="navigation">
          <NavLink
            to="/"
            end
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            }
          >
            Home
          </NavLink>

          <NavLink
            to="/lessons"
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            }
          >
            Lessons
          </NavLink>

          <NavLink
            to="/resources"
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            }
          >
            Resources
          </NavLink>

          <NavLink
            to="/about"
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            }
          >
            About
          </NavLink>
        </nav>
      </header>

```

```
<main className="site-main">
  <Outlet />
</main>

<footer className="site-footer">
  <p>© 2026 React Learning Studio. All rights reserved.</p>
</footer>
</div>
);
}
```

export default RootLayout;

Objašnjenje

RootLayout je komponenta koja se prikazuje na svim stranicama.

Ona sadrži:

header
navigation
main
footer

Najvažniji dio je:

```
<Outlet />
```

Outlet označava mjesto gdje će se prikazati trenutna child ruta.

Ako je korisnik na `/`, unutar **Outlet** komponente prikazuje se **Home**.

Ako je korisnik na `/lessons`, unutar **Outlet** komponente prikazuje se **Lessons**.

Ako je korisnik na `/resources`, prikazuje se **Resources**.

Ako je korisnik na `/about`, prikazuje se **About**.

Layout ostaje isti. Mijenja se samo sadržaj u **Outlet** dijelu.

Zašto koristimo **end** na Home linku?

Kod Home linka imamo:

```
<NavLink to="/" end>
```

end znači da Home link treba biti aktivan samo kada je URL točno `/`.

Bez `end`, Home link bi mogao ostati aktivan i na drugim rutama, jer sve rute počinju sa `/`.

6. `src/pages/Home.jsx`

```
import { Link } from "react-router-dom";

function Home() {
  return (
    <section className="page">
      <p className="eyebrow">Welcome</p>

      <h1>Learn React with a clear structure</h1>

      <p>
        React Learning Studio helps students understand React step by step.
        Lessons are organized around practical examples, clear explanations and
        small projects that build confidence.
      </p>

      <div className="home-actions">
        <Link className="button primary" to="/lessons">
          View Lessons
        </Link>

        <Link className="button secondary" to="/resources">
          Explore Resources
        </Link>
      </div>

      <div className="info-grid">
        <article className="info-card">
          <h2>Beginner friendly</h2>

          <p>
            Every topic starts from the basics. New concepts are introduced only
            when students have enough foundation to understand them.
          </p>
        </article>

        <article className="info-card">
          <h2>Practice focused</h2>

          <p>
            Students do not only read about React. They build small examples and
```

```
        complete exercises after important concepts.
      </p>
    </article>
  </div>
</section>
);
}
```

export default Home;

Objašnjenje

Home stranica koristi [Link](#) za navigaciju prema drugim stranicama:

```
<Link className="button primary" to="/lessons">
  View Lessons
</Link>
```

Za unutarnju navigaciju u React Router aplikaciji ne koristimo:

```
<a href="/lessons">View Lessons</a>
```

jer bi to izazvalo full page reload.

Koristimo [Link](#) zato što React Router promijeni URL bez ponovnog učitavanja stranice.

7. [src/pages/Lessons.jsx](#)

```
const lessons = [
  {
    id: 1,
    title: "Components and JSX",
    description:
      "Learn how React components return JSX and how JSX describes the user interface."
  },
  {
    id: 2,
    title: "Props and Children",
    description:
      "Practice passing data from parent components to child components using props."
  },
  {
    id: 3,
    title: "State and Events",
```

```

    description:
      "Understand how user actions update state and cause React to re-render the UI."
  },
  {
    id: 4,
    title: "Routing and Layouts",
    description:
      "Build multi-page single-page applications using routes, layout pages and Outlet."
  }
];

```

```

function Lessons() {
  return (
    <section className="page">
      <p className="eyebrow">Lessons</p>

      <h1>React lessons</h1>

      <p>
        This section contains a small learning path. Each lesson focuses on one
        important React concept and includes practical examples.
      </p>

      <div className="lesson-list">
        {lessons.map(lesson => (
          <article key={lesson.id} className="lesson-card">
            <span className="lesson-number">Lesson {lesson.id}</span>

            <h2>{lesson.title}</h2>

            <p>{lesson.description}</p>
          </article>
        ))}
      </div>
    </section>
  );
}

```

```
export default Lessons;
```

Objašnjenje

Ovdje koristimo array:

```
const lessons = [...]
```

i prikazujemo ga pomoću:

lessons.map(...)

Svaki element liste mora imati **key**:

```
<article key={lesson.id} className="lesson-card">
```

Ovo nije glavni fokus layout vježbe, ali daje stranici stvaran sadržaj.

8. `src/pages/Resources.jsx`

```
const resources = [
  {
    title: "Official Documentation",
    description:
      "Use documentation to check exact syntax, examples and recommended React patterns."
  },
  {
    title: "Practice Tasks",
    description:
      "Practice tasks help you turn theory into practical coding skills."
  },
  {
    title: "Project Ideas",
    description:
      "Small project ideas help you combine multiple concepts into real applications."
  }
];
```

```
function Resources() {
  return (
    <section className="page">
      <p className="eyebrow">Resources</p>

      <h1>Learning resources</h1>

      <p>
        Good learning does not stop after a lesson. Use resources, practice
        tasks and project ideas to strengthen your understanding.
      </p>

      <div className="resource-grid">
        {resources.map(resource => (
          <article key={resource.title} className="resource-card">
            <h2>{resource.title}</h2>
```

```

        <p>{resource.description}</p>
      </article>
    )}
  </div>
</section>
);
}

```

export default Resources;

Objašnjenje

Ova stranica također koristi `.map()` da bi prikazala više kartica.

Stranica nema svoj header, navigaciju ili footer. To je važno.

Te dijelove dobiva iz `RootLayout` komponente.

9. `src/pages/About.jsx`

```

function About() {
  return (
    <section className="page">
      <p className="eyebrow">About</p>

      <h1>How we teach React</h1>

      <p>
        React Learning Studio is built around one simple idea: students learn
        best when each concept is explained clearly, practiced immediately and
        connected to a real project.
      </p>

      <div className="about-section">
        <article className="info-card">
          <h2>Our teaching approach</h2>

          <p>
            We avoid jumping too quickly into advanced topics. Each lesson
            introduces one idea, explains why it matters and then shows how it
            works in code.
          </p>
        </article>
      </div>
    </section>
  );
}

```

```
<article className="info-card">
  <h2>Our principles</h2>

  <ul>
    <li>Clear explanations before complex patterns</li>
    <li>Practice after every important concept</li>
    <li>No skipped foundations</li>
    <li>Realistic examples instead of isolated snippets</li>
  </ul>
</article>
</div>
</section>
);
}
```

export default About;

Objašnjenje

About stranica sadrži samo sadržaj koji pripada About stranici.

Ne sadrži:

header
navigation
footer

To je ispravno, jer se ti elementi nalaze u layoutu.

10. **src/pages/NotFound.jsx**

```
import { Link } from "react-router-dom";

function NotFound() {
  return (
    <section className="page empty-state">
      <p className="eyebrow">404</p>

      <h1>Page not found</h1>

      <p>
        The page you are looking for does not exist. You can return to the home
        page and continue browsing.
      </p>
    </section>
  );
}
```

```
</p>

<Link className="button primary" to="/">
  Go to Home
</Link>
</section>
);
}

export default NotFound;
```

Objašnjenje

Ova stranica se prikazuje kada korisnik ode na rutu koja ne postoji.

Na primjer:

```
/something
/random-page
/test/abc
```

Zato u `App.jsx` imamo:

```
<Route path="*" element={<NotFound />} />
```

11. `src/App.css`

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: #eef2f7;
  color: #1f2937;
}
```

```
.site-shell {
  min-height: 100vh;
  display: flex;
  flex-direction: column;
}
```

```
.site-header {
  max-width: 1040px;
  width: 100%;
  margin: 0 auto;
  padding: 20px 24px;
```

```
box-sizing: border-box;
```

```
display: flex;  
align-items: center;  
justify-content: space-between;  
}
```

```
.brand {  
font-size: 22px;  
font-weight: bold;  
}
```

```
.navigation {  
display: flex;  
gap: 12px;  
flex-wrap: wrap;  
}
```

```
.nav-link {  
padding: 8px 12px;  
color: #344054;  
text-decoration: none;  
border-radius: 8px;  
}
```

```
.nav-link:hover {  
background-color: #e9eef6;  
}
```

```
.nav-link.active {  
background-color: #1f2937;  
color: white;  
}
```

```
.site-main {  
max-width: 1040px;  
width: 100%;  
margin: 0 auto;  
padding: 24px;  
box-sizing: border-box;  
flex: 1;  
}
```

```
.site-footer {  
max-width: 1040px;  
width: 100%;  
margin: 0 auto;  
padding: 20px 24px;
```

```
box-sizing: border-box;
color: #667085;
}
```

```
.page {
background-color: white;
border: 1px solid #d8dee9;
border-radius: 16px;
padding: 32px;
}
```

```
.eyebrow {
margin: 0 0 8px;
color: #475467;
text-transform: uppercase;
letter-spacing: 0.08em;
font-size: 13px;
font-weight: bold;
}
```

```
.page h1 {
margin-top: 0;
font-size: 36px;
}
```

```
.page p {
line-height: 1.6;
}
```

```
.home-actions {
display: flex;
gap: 12px;
flex-wrap: wrap;
margin-top: 24px;
}
```

```
.button {
display: inline-block;
padding: 10px 16px;
border-radius: 8px;
text-decoration: none;
border: 1px solid #ccd3df;
}
```

```
.button.primary {
background-color: #1f2937;
color: white;
}
```

```
.button.secondary {  
  background-color: white;  
  color: #1f2937;  
}
```

```
.info-grid,  
.lesson-list,  
.resource-grid,  
.about-section {  
  display: grid;  
  gap: 20px;  
  margin-top: 24px;  
}
```

```
.info-grid,  
.about-section {  
  grid-template-columns: repeat(2, 1fr);  
}
```

```
.lesson-list {  
  grid-template-columns: repeat(2, 1fr);  
}
```

```
.resource-grid {  
  grid-template-columns: repeat(3, 1fr);  
}
```

```
.info-card,  
.lesson-card,  
.resource-card {  
  border: 1px solid #d8dee9;  
  border-radius: 12px;  
  padding: 20px;  
  background-color: #f8fafc;  
}
```

```
.info-card h2,  
.lesson-card h2,  
.resource-card h2 {  
  margin-top: 0;  
}
```

```
.lesson-number {  
  display: inline-block;  
  margin-bottom: 10px;  
  padding: 4px 8px;  
  border-radius: 999px;
```

```
background-color: #e9eef6;
color: #344054;
font-size: 13px;
font-weight: bold;
}

.empty-state {
text-align: center;
}

@media (max-width: 760px) {
.site-header {
align-items: flex-start;
flex-direction: column;
gap: 16px;
}

.info-grid,
.lesson-list,
.resource-grid,
.about-section {
grid-template-columns: 1fr;
}

.page h1 {
font-size: 30px;
}
}
```

12. Kako testirati rješenje

Pokrenite aplikaciju:

```
npm run dev
```

Otvorite aplikaciju u browseru.

Testirajte ove URL-ove:

```
/
```

Treba se prikazati Home stranica.

```
/lessons
```

Treba se prikazati Lessons stranica.

/resources

Treba se prikazati Resources stranica.

/about

Treba se prikazati About stranica.

/something-random

Treba se prikazati NotFound stranica.

13. Što se mora dogoditi u browseru

Kada prelazite s jedne stranice na drugu:

Home → Lessons → Resources → About

stranica se ne smije potpuno reloadati.

Header mora ostati na vrhu.

Footer mora ostati na dnu.

Sadržaj u sredini se mijenja.

Aktivni link u navigaciji mora imati drugačiji stil.

To pokazuje da `RootLayout` i `Outlet` rade ispravno.

14. Najvažniji dio rješenja

Najvažniji dio cijele vježbe je ovaj odnos:

```
<Route path="/" element={<RootLayout />}>
  <Route index element={<Home />} />
  <Route path="lessons" element={<Lessons />} />
  <Route path="resources" element={<Resources />} />
  <Route path="about" element={<About />} />
  <Route path="*" element={<NotFound />} />
</Route>
```

i ovaj dio u layoutu:

```
<main className="site-main">  
  <Outlet />  
</main>
```

Ovo znači:

RootLayout se prikazuje uvijek.

Home, Lessons, Resources, About i NotFound se mijenjaju unutar Outlet-a.

15. Kratko objašnjenje za polaznike

Layout stranice koristimo zato da ne ponavljamo iste dijelove aplikacije na svakoj stranici.

Bez layouta bismo morali header, navigaciju i footer kopirati u svaku page komponentu.

S layoutom to pišemo samo jednom:

RootLayout.jsx

A svaka stranica sadrži samo svoj specifični sadržaj.

Outlet je mjesto gdje React Router ubacuje aktivnu child rutu.

Zato je mentalni model vrlo jednostavan:

Layout ostaje isti.

Ruta odlučuje koja stranica je aktivna.

Outlet prikazuje aktivnu stranicu.

To je temelj organizacije većih React Router aplikacija.